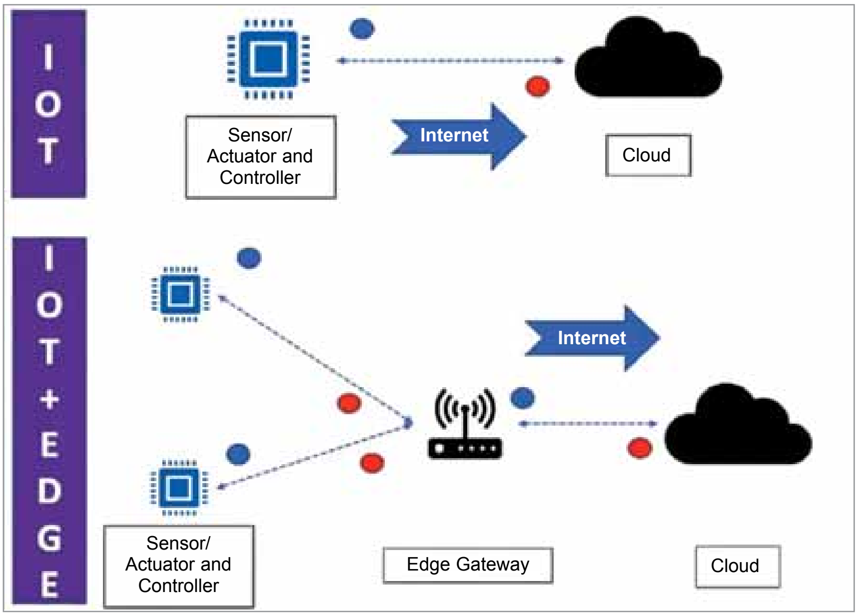


*IoT in a nutshell*

*Apollo simulators at mission control in Houston (Credit: NASA)*

chosen astronauts to face any emergency. On the day of the big event, the astronauts aboard Apollo 13 had access to only its telemetry data. Fortunately, the same telemetry data was accessible by the ground station on the Earth as well. Despite being far apart by several thousand kilometres, communication was established wherein exchange of messages took place thanks to virtual representation. Before giving various instructions, verification was done. If the results were satisfactory, only then communication ensued. This constant contact played a big role in bringing the Apollo 13 spacecraft safely back to Earth.

Since there was no use of IoT, it could not technically be called a Digital Twin. But it qualifies as an example of Digital Twin as it combined multiple things for communication purposes. This made the whole system capable of multitasking and responsive.

In 1977, several flight simulators were introduced, which allowed pilots to practice their flight techniques. In 1982, AutoCAD was launched, which eased the design of 2D and 3D models in the times to come.

In 2002, Dr Grieves introduced the first concept of Digital Twin and in 2011, NASA published several papers on the subject. In 2015, GE became the first company to take the Digital Twin initiative. After 2015, several industrial organisations started adopting Digital Twin as part of their operations.

## IoT and its shortcomings

As we all know, IoT stands for the Internet of Things. Since we humans have been using the Internet for a long time to communicate with each other, the term IoT is also referred to as the Internet of People. The word 'things' generally refers to physical objects such as a lightbulb, refrigerator, or a machine in a factory. Anything that can communicate with other entities via the Internet can be a part of the Internet of Things.

IoT comprises sensors, actuators, and controllers. Sensors are electromechanical or electrochemical devices, which sense physical entities and convert them into electrical signals.

Actuators are devices that receive those electrical signals and perform a physical action. And finally, the controller, which acts as the brain, controls both sensors and actuators. It provides electrical signals to the actuator and reads electrical signals from the sensor.

When IoT devices communicate information in conjunction, the data gets stored in the cloud. IoT architecture may look simple but it has several drawbacks. Absence of the Internet connection, for instance, can lead to catastrophic failures as the sensor and the actuator won't be able to communicate with the cloud, causing data loss.

Also, if every individual sensor tries to communicate with the cloud at the same time, the transmission of many packets will interfere or slow down the path from the sensor or controller to the cloud. To avoid such scenarios, edge computing is used.

When edge computing and IoT work together, the sensor and controller do not talk to the cloud directly. Instead, communication takes place through an edge gateway, which aggregates all the data and transfers it to the cloud. Here, no Internet connection is required between the sensor/actuator and the controller and the edge gateway.

This way, response is obtained even when there is no Internet connection. This can reduce the latency and the number of messages being sent to the cloud. The edge gateway can process data locally and send only important and meaningful information to the cloud.

## Where Digital Twin meets IoT

Digital Twin without IoT is nothing but a static simulation model. With the addition of Digital Twin we can get real-time data from physical things via IoT. That makes the entire thing a living-learning model.